

Q1

The Chromakey failure in Image B is due to improper lighting or color spill. If the background (green screen) has uneven lighting or shadows, the Chromakey cannot cleanly separate the foreground. However, foreground objects with colors similar to the background may not be detected correctly, causing parts of the background to remain visible.

Q2

~~The triple image~~.

The triple difference image is option C. Triple differencing subtracts consecutive frame differences, reducing noise and highlighting true motion. Option C shows the man's movement clearly, while others retain static artifacts or noise.

Thus, three frames: I_1 , I_2 and I_3 , the triple difference image is given by:

$$D = |(I_3 - I_2)| + |(I_2 - I_1)|$$

(Q3)

Effect of erosion: Removes boundary pixels, shrinking objects and eliminating small details.

Eroded image (applied with structural element): let's assume a 3×3 square structuring element, the processed image would have thinner regions, with isolated pixels removed.

Also, we can shrink bright regions and enlarge ~~large~~ dark regions.

(Q4)

Effect of Dilation: Dilation is the opposite of erosion. It expands objects and fills small holes.

Hence, it enhances bright regions, it thickens edges and finally it connects broken components.

It also adds pixels to object boundaries, enlarging regions.

Dilated image (using the vertical structuring element): pixels are extended vertically, filling gaps in columns per the 5-row vertical kernel.

(Q7) ~~(Q8)~~

(4)

let's compute D_4 , D_8 and D_m distances for two pixels $a(3,0)$ and $b(0,3)$:

① D_4 (City Block / Manhattan Distance) or D_4 (Lawrence Distance):

$$D_4(a, b) = |x_2 - x_1| + |y_2 - y_1|$$

$$D_4(3, 0 \rightarrow 0, 3) = |0 - 3| + |3 - 0| = 3 + 3$$

$$D_4 = 6$$

② D_8 (Chessboard Distance):

$$D_8(a, b) = \max(|x_2 - x_1|, |y_2 - y_1|)$$

$$~~D_8(3, 0)~~ \quad D_8(3, 0 \rightarrow 0, 3) = \max(3, 3)$$

$$D_8 = 3$$

③ D_m (Euclidean Distance) or D_m (Path Distance):

$$D_m(a, b) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\begin{aligned} D_m(3, 0 \rightarrow 0, 3) &= \sqrt{(3-0)^2 + (0-3)^2} \\ &= \sqrt{9 + 9} = \sqrt{18} \end{aligned}$$

$$D_m = 4.24$$

~~8-adjacents~~

Q6

5

8-adjacencies in $\sqrt{\{1:11\}}$

$(0,1) = 10$: Adjacent to $(1,0) = 8$ (Value 8), $(1,1) = 130$ (excluded)

$(2,0) = 11$: Adjacent to $(3,0) = 8$, $(2,1) = 43$ (excluded)

$(3,3) = 7$: Adjacent to $(3,2) = 121$ (excluded),
 $(2,3) = 25$

Thus, Marked adjacencies: $(10 \leftrightarrow 8)$, $(11 \leftrightarrow 8)$, $(7 \leftrightarrow 25)$,
etc. Because 8-adjacency considers horizontal, vertical, and
diagonal neighbors.

Similarly, The 8-adjacencies in $\sqrt{\{1:11\}}$ are:

1. $(0,3) = 8$ is 8-adjacent to $(1,2) = 2$
2. $(1,2) = 2$ is 8-adjacent to $(0,3) = 8$ and $(0,1) = 10$
3. $(2,0) = 11$ $\swarrow \searrow$ to $(3,0) = 8$
4. $(3,0) = 8$ $\swarrow \searrow$ to $(2,0) = 11$ and $(3,1) = 7$
5. $(3,1) = 7$ $\swarrow \searrow$ to $(2,0) = 11$ and $(3,0) = 8$